

Student Placement Predictor

Project Overview

The **Student Placement Predictor** is a Machine Learning project that predicts whether a student is likely to be placed based on academic performance and aptitude-related features. The project uses classification algorithms to analyze student data and generate placement predictions.

Features

- Predicts whether a student will be **Placed** or **Not Placed**.
 - Performs data preprocessing before model training.
 - Splits the dataset into training and testing sets.
 - Compares multiple Machine Learning algorithms.
 - Saves the best-performing model using Pickle for future use.
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Technologies Used

- Python
 - NumPy
 - Pandas
 - Scikit-learn
 - Pickle
 - Jupyter Notebook
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Project Structure

```
Student-Placement-Predictor/  
|  
├── student-placement-predictor.ipynb    # Main notebook  
├── students_placement.csv               # Dataset  
├── model.pkl                           # Trained model  
└── README.md                           # Project documentation
```

Machine Learning Workflow

1. Import required libraries.

2. Load the placement dataset.
3. Explore and inspect the data.
4. Separate features and target variable.
5. Split the dataset into training and testing sets.
6. Standardize the feature values.
7. Train multiple classification models:
8. Logistic Regression
9. Random Forest Classifier
10. Support Vector Machine (SVM)
11. Evaluate model performance using Accuracy Score.
12. Select the best-performing model.
13. Save the trained model using Pickle.



Algorithms Used

- Logistic Regression
- Random Forest Classifier
- Support Vector Machine (SVM)



Required Libraries

Install the required packages using:

```
pip install numpy pandas scikit-learn
```



How to Run

1. Clone the repository.

```
git clone https://github.com/your-username/Student-Placement-Predictor.git
```

1. Navigate to the project folder.

```
cd Student-Placement-Predictor
```

1. Install the required libraries.

```
pip install numpy pandas scikit-learn
```

1. Open the notebook.

```
jupyter notebook
```

1. Run all cells to train the model and generate predictions.

Model Output

The trained model predicts:

- **Placed (1)** – Student is likely to get placed.
- **Not Placed (0)** – Student is less likely to get placed.

Model Saving

The trained model is saved using Pickle.

```
import pickle

pickle.dump(model, open("model.pkl", "wb"))
```

The saved model can later be loaded for deployment or prediction.

Future Improvements

- Build a web application using Flask or Streamlit.
- Hyperparameter tuning for better accuracy.
- Add more student performance features.
- Deploy the model on cloud platforms.
- Improve visualization and model comparison.

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★ Acknowledgement

This project was developed for learning Machine Learning concepts, including data preprocessing, classification algorithms, model evaluation, and deployment using Python and Scikit-learn.